

Assignment 8 – Invertebrate taxonomic richness in glacier-fed streams at different latitudes

1. The hypothesis is concerned with the effects of latitude and environmental stress on the dependent variable, taxon richness. Therefore, the model will contain latitude as a continuous independent variable and environmental stress as a three level categorical independent variable. The authors will also be concerned with the effect of an interaction between latitude and environmental stress. There are continuous and categorical independent variables so ANCOVA will be used to analyse the full model with interactions. All factors will be tested for an effect in both direction. The authors may predict a direction due to the hypothesis however, an effect opposite of the predicted would still be interesting. If the full model finds interactions to be nonsignificant they will be removed from the model. The authors are interested in both the main effects of latitude and environmental stress therefore, they will remain in the model even if insignificant.

2.

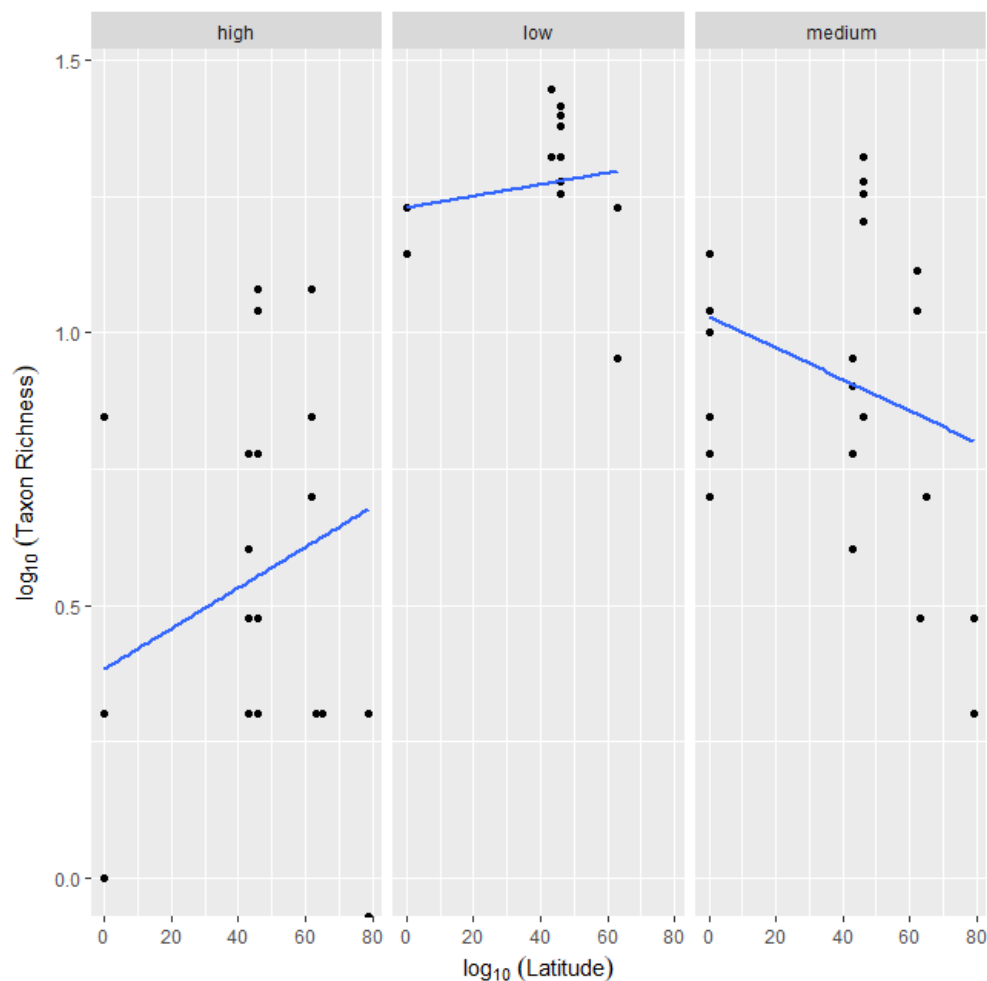


Figure 1: Scatter plots of the effect of latitude on taxon richness at each level of environmental stress.

3.

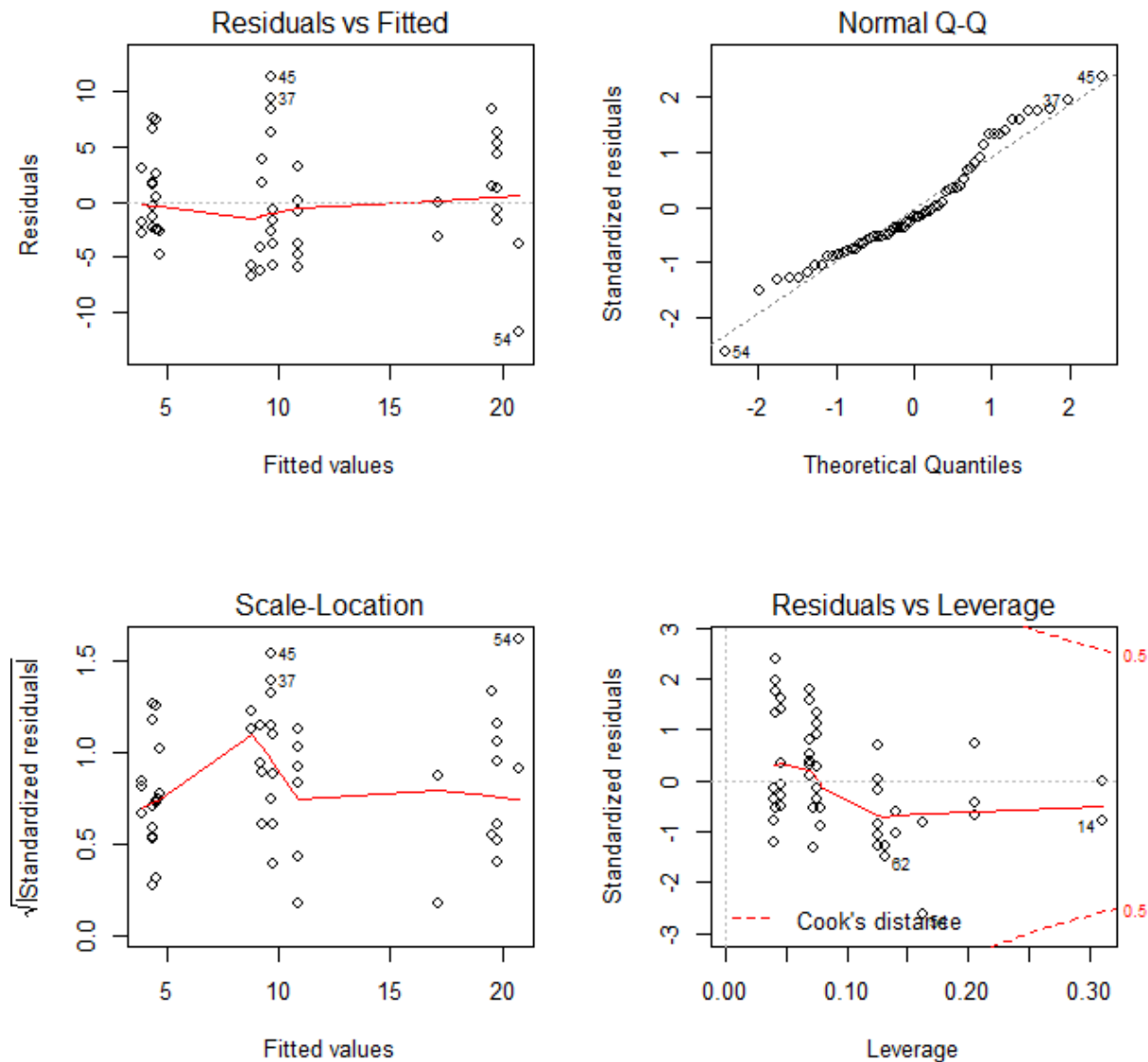


Figure 2: Residual diagnostic plots

Residuals vs. Fitted plot is straight showing the data is homoscedastic. The Normal Q-Q shows deviations from normality however the sample size is large and therefore robust. In the Residuals vs. Leverage plot no data points are outside Cook's distance showing there is no effect of leverage on the linear model. The data looks to be homoscedastic and follow a normal distribution with no effect of outliers. The sample size is large making the model robust to small violations. Therefore, the statistical assumptions are met and analyses of the data can continue.

4.

Full model with interactions

```
> model.full<-lm(Taxon.richness ~ Glacier.Index.cat + Latitude.N.S + Glacier.Index.cat:Latitude.N.S, data = Glaciers)
> Anova(model.full, type = 3)
Anova Table (Type III tests)

Response: Taxon.richness

              Sum Sq Df F value    Pr(>F)
(Intercept)      74.06  1   3.1253 0.082439 .
Glacier.Index.cat 349.63  2   7.3769 0.001415 **
Latitude.N.S       1.32  1   0.0556 0.814492
Glacier.Index.cat:Latitude.N.S  33.03  2   0.6970 0.502281
Residuals      1350.75 57
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The main effect of latitude was found to be insignificant with p-value = 0.814. The main effect of environmental harshness, Glacier.Index.cat, was found to be significant with p-value = 0.001415. The effect of an interaction between Glacier.Index.cat was found to be insignificant with p-value = 0.502. The effect of the interaction was found to be insignificant and therefore an analysis should be done without the effect of interaction.

Model without interactions

```
> model.ancova<-lm(Taxon.richness ~ Glacier.Index.cat + Latitude.N.S, data = Glaciers)
> Anova(model.ancova, type = 3)
Anova Table (Type III tests)

Response: Taxon.richness

              Sum Sq Df F value    Pr(>F)
(Intercept)      184.92  1   7.8845 0.006749 **
Glacier.Index.cat 1989.04  2  42.4032 3.852e-12 ***
Latitude.N.S        0.03  1   0.0012 0.972035
Residuals      1383.78 59
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

When the interaction was removed from the model the main effect of latitude remained insignificant and the p-value increased to 0.972. The main effect of Glacier.Index.cat remained significant and the p-value decreased to 3.852×10^{-12} . As a result of the differences between the full model and the partial model, when looking at the effect of interactions one should look at the full model, when looking at the main effects one should look at the partial model.

5.

Summary of model with interactions

```
> summary(model.full)

Call:
lm(formula = Taxon.richness ~ Glacier.Index.cat + Latitude.N.S +
    Glacier.Index.cat:Latitude.N.S, data = Glaciers)

Residuals:
    Min       1Q   Median       3Q      Max
-11.7276  -2.9041  -0.8551   2.7767  11.3373

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      3.90413    2.20841   1.768 0.082439 .
Glacier.Index.catlow    13.21362    3.49711   3.778 0.000379 ***
Glacier.Index.catmedium    6.95099    2.80519   2.478 0.016202 *
Latitude.N.S         0.01029    0.04367   0.236 0.814492
Glacier.Index.catlow:Latitude.N.S    0.04701    0.07595   0.619 0.538450
Glacier.Index.catmedium:Latitude.N.S -0.03622    0.05728  -0.632 0.529707
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.868 on 57 degrees of freedom
Multiple R-squared:  0.6013,    Adjusted R-squared:  0.5663
F-statistic: 17.19 on 5 and 57 DF,  p-value: 2.448e-10
```

Summary of model without interactions

```
> summary(model.ancova)

Call:
lm(formula = Taxon.richness ~ Glacier.Index.cat + Latitude.N.S,
    data = Glaciers)

Residuals:
    Min       1Q   Median       3Q      Max
-10.355  -2.832  -1.340   2.649  11.148

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      4.323e+00  1.540e+00   2.808 0.006749 **
Glacier.Index.catlow    1.498e+01  1.629e+00   9.194 5.44e-13 ***
Glacier.Index.catmedium    5.488e+00  1.411e+00   3.891 0.000257 ***
Latitude.N.S         9.009e-04  2.559e-02   0.035 0.972035
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The effects of interaction between latitude and environmental harshness were not significant as shown in the summary above. The interaction between Latitude.N.S and Glacier.Index.catlow had a slope of

0.0573 with a standard error of 0.07595. The interaction between Latitude.N.S and Glacier.Index.catmedium had a slope of -0.02593 with a standard error of 0.05728. The interaction between Latitude.N.S and Glacier.Index.cathigh had a slope of 0.01029 with a standard error of 0.04367. A positive slope means that in the given level of environmental harshness the taxon richness increases with latitude. That means that taxon richness increases with increasing latitude in low and high environmental harshness but decreases in medium environmental harshness. These findings are in disagreement with the biological hypothesis however the slope values are very small and the standard errors are larger than the slopes showing that any pattern could be observed. The results were found as not statistically significant however they may still be biologically significant. The results disagree with the biological hypothesis and therefore create more questions as to why this observed pattern of taxon richness is occurring.

The interactions in the full model were found to be not significant and were therefore, removed from the model to look at the main effects of environmental harshness and latitude. The mean number of species in Glacier.Index.catlow streams is 19.303 ± 1.629 . The mean number of species in Glacier.Index.catmedium streams is 9.811 ± 1.411 . The mean number of species in Glacier.Index.cathigh streams is 4.323 ± 1.540 . Therefore, the range in taxon difference from low environmental harshness to high environmental harshness is 14.98 species and agrees with the biological hypothesis that taxon richness decreases with increasing environmental harshness. A change of 14.98 species is equivalent to a 77.6% decrease in taxon richness. The effect of environmental harshness is biologically significant.

The main effect of latitude had a slope of 0.0009004 with a standard error of 0.02559. The range of the slope of the main effect of latitude extends from -0.02468 to 0.02649. This means over a range of latitudes from 0 to 79 degrees the change in taxon richness is equal to 0.0711 ± 2.02161 . This range includes negative values which agree with the biological hypothesis but also contains positive values and 0 which disagree with the biological hypothesis. A maximum increase or decrease of approximately only 2 species is equal to a 10.4% change in taxon richness. This change in taxon richness isn't very large for this particular environment, however, if this pattern is consistent across many environments or even the globe then a 10% change in taxon richness is biologically significant.

6.

The assumptions of the general linear model are met with satisfaction. Although the sample size is large many components of the model were found to be not significant with standard errors that are larger than the estimated value leading me to believe power may be an issue. The experimental design is good however, I feel it is limited in the ability to extrapolate as the study is restricted to glacier streams and macroinvertebrates.

